Institute of Technology of Cambodia
Faculty of Electrical Engineering
Génie Informatique et Communication

GICI5CC · CLOUD COMPUTING · WEEK 13

AI/ML Services in the Cloud

Machine learning pipelines, pre-trained AI services, Amazon SageMaker and model deployment.



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Recap

LAST SESSION

DevOps and CI/CD

Pipelines that build, test and deploy our code automatically.

LAST SESSION

Infrastructure as Code

Infrastructure defined as configurations, kept in source control.

We can now deploy an application reliably. Today we make it smarter.

Learning objectives

1. Explain what machine learning is, and the steps of an ML pipeline.
2. Describe the pre-trained AI services on AWS.
3. Give an overview of Amazon SageMaker.
4. Compare the ways to deploy an ML model.
5. Match AI/ML services to use cases.

THIS WEEK

Topic 12

AI/ML services in the cloud. The capstone project moves into development (CLO2, CLO3).

DISCUSSION

Where did you use AI today?

Unlocking a phone with your face, a translated message, a recommended video, a chatbot answer.

PART 1

Machine learning in one session

Enough of the ideas to use the cloud services well.

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AI, machine learning, deep learning, generative AI

01

Artificial intelligence

Any technique that lets computers do tasks that normally need human intelligence.

02

Machine learning

A part of AI. The computer learns patterns from data instead of following rules we write.

03

Deep learning

A part of ML. It uses neural networks with many layers, and suits images, speech and text.

04

Generative AI

Deep learning models that create new text, images or code, built on large foundation models.

Each one sits inside the one before it.

Traditional programming vs machine learning

RULES + DATA = ANSWERS

Traditional programming

We write the rules. The program applies them to data and produces answers. It works when we can describe the rules, such as calculating a bill.

DATA + ANSWERS = RULES

Machine learning

We provide data and the answers we expect. Training produces the rules, called a model. It works when the rules are too complex to write, such as recognizing a face.

Three types of machine learning

Supervised

Learns from labeled examples: inputs with the correct answer. Used to predict a price, or classify an email as spam or not.

Unsupervised

Finds structure in data that has no labels. Used to group similar customers, or spot unusual transactions.

Reinforcement

Learns by trial and error, from rewards and penalties. Used for robots, games and control systems.

Training and inference

LEARN ONCE

Training

Showing the algorithm a large amount of data so it learns a model. Heavy computation, often on GPUs, done occasionally.

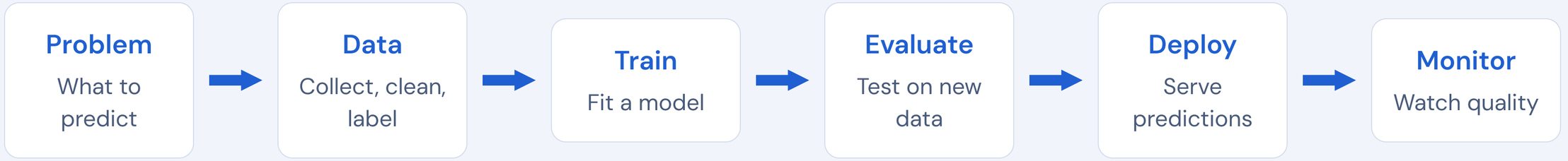
USE MANY TIMES

Inference

Using the trained model to make a prediction on new data. Light computation, done every time the application asks.

The two have different cost patterns, so the cloud offers different services for each.

The ML pipeline



A pipeline is not finished at deployment. The world changes, the model's quality drifts, and it is trained again on newer data.

Why AI/ML runs in the cloud



Compute on demand

Training needs powerful machines for hours, not years. We rent GPUs and release them.



Data at scale

Models need large datasets. Cloud storage such as Amazon S3 holds them cheaply and durably.



Managed services

The provider runs the servers and frameworks, so a small team can ship ML features.



Pay for use

No upfront hardware. We pay while training runs and while the model serves requests.

PART 2

AI/ML services on AWS

Three layers: the higher we go, the less machine learning we need to know.

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The AWS AI/ML stack

Layer	What we get	Who it is for	Examples
AI services	Pre-trained models behind an API	Any developer	Amazon Rekognition, Amazon Polly, Amazon Bedrock
ML services	A managed platform to build our own models	Data scientists and ML engineers	Amazon SageMaker
Frameworks and infrastructure	Raw compute and ML libraries	ML experts who want full control	PyTorch or TensorFlow on Amazon EC2 GPU instances

It mirrors the service models from Week 2: more managed at the top, more control at the bottom.

Pre-trained AI services

Service	What it does	Example use
Amazon Rekognition	Analyzes images and video	Detect objects, faces and unsafe content
Amazon Textract	Extracts text and data from documents	Read scanned forms and invoices
Amazon Comprehend	Understands text	Find the sentiment of customer reviews
Amazon Translate	Translates text between languages	Localize a website
Amazon Transcribe	Turns speech into text	Subtitle a recorded lecture
Amazon Polly	Turns text into speech	Read an article aloud
Amazon Lex	Builds chatbots and voice bots	A customer support assistant

Using a pre-trained service

```
import boto3

client = boto3.client("rekognition")
response = client.detect_labels(
    Image={"S3Object": {"Bucket": "my-bucket",
                        "Name": "photo.jpg"}},
    MaxLabels=5)

for label in response["Labels"]:
    print(label["Name"], label["Confidence"])
```

No model to train, and no server to run. We send a request and get a result.

Here, Python asks Amazon Rekognition what is in a photo stored in Amazon S3, and prints each label with its confidence.

We pay for each request.

Generative AI: Amazon Bedrock

Foundation models from several providers, behind *one API*.

What it is

A fully managed service for building generative AI applications, with no infrastructure to manage.

What we do with it

Generate and summarize text, answer questions, build chat assistants, create images.

Our own data

A model can be given our documents to answer from, without training a new model.

PART 3

Amazon SageMaker

When no pre-trained service fits, we build, train and deploy our own model.

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SageMaker overview

A fully managed service to *build, train and deploy* machine learning models.

01

Build

Notebooks in the browser to explore data and write the model code. Built-in algorithms and ready-made models to start from.

02

Train

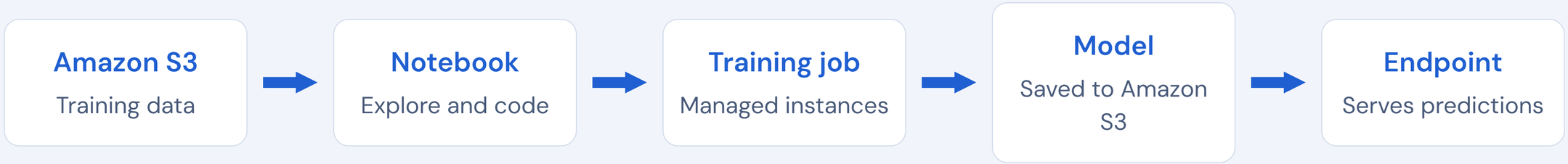
Training jobs run on machines that SageMaker starts, uses and shuts down. We pay only while the job runs.

03

Deploy

The trained model is hosted behind an endpoint that applications call for predictions.

A model’s journey through SageMaker



Data and the trained model both live in Amazon S3. Compute appears only for the steps that need it.

ML model deployment

Option	How it works	Best for
Real-time endpoint	Always running, answers in milliseconds	Interactive applications with steady traffic
Serverless inference	Starts on request and scales to zero when idle	Occasional or unpredictable traffic
Asynchronous inference	Queues requests and returns results later	Large inputs and long processing times
Batch transform	Runs once over a whole dataset, then stops	Predictions for many records at once, with no live users

Ask two questions: does someone wait for the answer, and how often do requests arrive?

Deployment and cost

Endpoints cost while they run

A real-time endpoint is billed for every hour it exists, even with no requests. Delete it when the demo is over.

GPUs are expensive

Training and serving on GPU instances costs far more than the instances from our labs. Start small.

Check your Learner Lab

Not every AI/ML service is available in every lab account. Check what yours allows before you design around it.

PART 4

Choosing and using AI/ML

Start with the simplest service that solves the problem.

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Which service should I use?

01

TRY THIS FIRST

A pre-trained AI service

The task is common: reading documents, analyzing images, translating, speech. No ML expertise needed.

02

FOR GENERATIVE TASKS

Amazon Bedrock

The task is generating or understanding language or images, with a foundation model.

03

WHEN NOTHING ELSE FITS

Amazon SageMaker

The problem is specific to our own data, and no existing model solves it well.

The further right, the more data, skill, time and cost it takes.

AI/ML use cases

Document processing

Extract the fields from invoices and forms automatically. Amazon Textract.

Content moderation

Flag unsafe images before they are published. Amazon Rekognition.

Customer support

A chatbot that answers common questions at any hour. Amazon Lex or Amazon Bedrock.

Review analysis

Measure whether customer feedback is positive or negative. Amazon Comprehend.

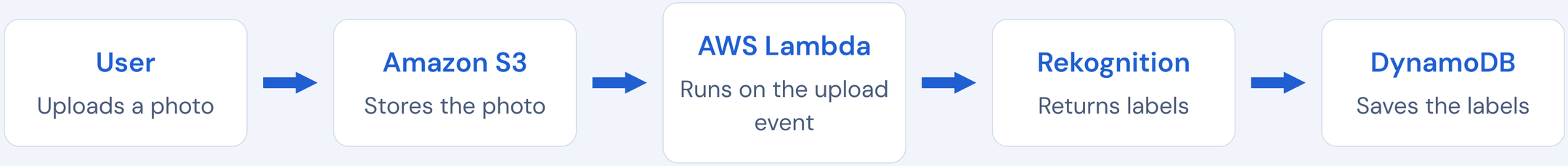
Accessibility

Subtitles for videos and audio for articles. Amazon Transcribe and Amazon Polly.

Custom prediction

Predict demand or detect fraud from the company’s own records. Amazon SageMaker.

Example: adding AI to an application



Every piece is a service from this course: storage, serverless, an event, a database, and one API call for the AI.

Using AI responsibly



Bias

A model learns from its data. If the data is unfair or incomplete, the predictions will be too.



Privacy

Know what personal data you send to a service, and whether you are allowed to.



Accuracy

Models make mistakes, and generative models can state wrong things confidently. Check the output.



Responsibility

The shared responsibility model still applies: the provider secures the service, we answer for how we use it.

Capstone project: development

1. Explore the AI/ML services, and decide whether one belongs in your project.
2. Begin implementation from your proposal and architecture design.
3. Implement the core features first.
4. Keep the weekly project update going.

COURSE POLICY

Remember

Three weeks with no project update means zero for the poster presentation.

Key takeaways

01

ML learns from data

A model is trained once on data, then used many times for inference.

02

Start with an API

Pre-trained AI services add vision, speech and language with one request.

03

SageMaker for your own model

A managed way to build, train and deploy when nothing existing fits.

04

Deployment is a choice

Real-time, serverless, asynchronous or batch, chosen by traffic and by cost.

QUIZ

Post-knowledge quiz

ML pipelines, pre-trained AI services, SageMaker and model deployment.

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Next session



01

How to choose a cloud platform



02

Start your cloud journey



03

Project development